IIM Sirmaur Smart Campus 2035: The Campus Operations Control Room

Operations Management Course Project (BBA), Project 4. Draft working document.

1. The Idea in One Page

Most "smart campus" proposals are a shopping list: AI classrooms, IoT sensors, a digital twin. Our proposal is different. We propose **one operational brain for the whole campus: the Campus Operations Control Room.**

The Control Room is a single data layer and dashboard that collects what is happening on campus (meals eaten, complaints raised, electricity and water used, shuttle requests, the academic calendar), forecasts what will happen next, and then **recommends or triggers operational decisions**. It does not just show charts. It tells the mess how much to cook tomorrow, tells the estates team which asset to replace instead of repair, and tells the Director when the campus will outgrow its current capacity.

Three principles shape the design:

1. **Decisions, not dashboards.** Every screen ends in an action: cook this quantity, send this technician, run this shuttle, start this expansion.
2. **One demand signal drives everything.** The academic calendar (exams, placements, fests, breaks) is the master input for every forecast, so the modules talk to each other instead of working in silos.
3. **Built for Sirmaur, not for a brochure.** Monsoon road closures, power cuts, fog, and patchy connectivity are part of the design through an offline-first "Resilience Mode".

2. The Problem

IIM Sirmaur is a young institute in a remote location, and it is growing. Its operations were set up for a small community and are now stretched in predictable ways.

Area	Typical problem today
Mess	Over- or under-production, food waste, peak-hour queues, no link between menu, headcount, and procurement

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Hostel and facilities	Complaints through scattered channels (calls, messages, in person), no tracking, no priority rules, repeat breakdowns of the same assets
Energy and water	Consumption is invisible. Nobody can say which building wastes what, or spot a leak or a runaway load early
Transport	Fixed schedules that do not match when students actually need to travel to and from town
Student services	Students chase offices for answers that already exist in circulars and FAQs
Planning	Capacity decisions (mess seats, beds, water, power) are made reactively, without a link to intake growth
Resilience	Disruptions (landslides, power cuts, connectivity loss) are handled ad hoc

Root cause (the OM view): these are not seven separate problems. They share one cause: *the campus operates without a common demand forecast and without a feedback loop between demand, capacity, and service performance.* Each department reacts to what already happened. The Control Room closes that loop.

(Note for the team: we should validate each row above with our own observations, interviews, and surveys at Paonta Sahib before the report states it as fact.)

3. Features and How Each Solves the Problem

The Control Room has eight screens. Each one maps to a problem above, a decision it supports, and an OM method behind it.

3.1 Command Home

A one-page view of today's recommended decisions, key KPIs, and alerts, with a **Normal / Resilience Mode** toggle. Resilience Mode switches the rules: larger stock buffers, offline ticketing, rerouted transport.

3.2 Demand and Mess

- **What it does:** Forecasts meal-wise headcount, recommends cook quantities, generates the ordering list, and tracks waste.
- **Method:** Moving average and exponential smoothing, a day-of-week index, and an event factor from the calendar. Cook quantity is the forecast plus a safety buffer. Perishables use newsvendor logic (balance the cost of running out against the cost of leftovers). Dry stock uses a reorder point (average daily use × lead time + safety stock). Forecast quality is tracked with MAD or MAPE.
- **Circular local loop:** Sourcing produce from nearby Sirmaur farmers and vendors, and routing mess waste to compost or biogas. Tracked as kg of waste diverted and food-miles saved.
- **Problem solved:** Waste, shortages, queues, and unplanned procurement.

3.3 Maintenance and Helpdesk

- **What it does:** One ticketing channel with priority rules, technician scheduling, a preventive maintenance calendar, and a repair-or-replace recommendation.
- **Method:** Pareto analysis to find the few assets causing most complaints, MTBF (mean time between failures) per asset type, a repair-vs-replace rule (replace when yearly repair cost exceeds the annualized cost of a new unit, or after N failures in M months), and queue utilization ($\rho = \lambda / (c \times \mu)$) to size the technician team.
- **Problem solved:** Untracked complaints and repeat breakdowns.

3.4 Energy and Water

- **What it does:** Sub-metered consumption by building against a per-student baseline, with anomaly alerts, a solar payback case, and a rainwater harvesting estimate.
- **Method:** Baseline per student, control-chart alerts (mean + 2 standard deviations), solar payback (investment ÷ annual savings), rainwater potential (roof area × rainfall × runoff coefficient), and carbon (kWh × grid emission factor).
- **Problem solved:** Invisible consumption, leaks, and wasted load.

3.5 Transport

- **What it does:** Shows travel demand by hour and day and proposes a shuttle plan, with seat booking.

- **Method:** Demand histogram from the survey, trips needed = ceiling of peak demand ÷ seats per vehicle, and a fixed-vs-on-demand comparison on utilization, wait time, and cost per trip.
- **Problem solved:** Schedules that do not match real demand.

3.6 What-If Simulator (our "digital twin")

Instead of a 3D model, we build a **flow simulation** of how students move between hostel, mess, classes, and library. Managers can test decisions before making them: stagger mess slots, grow the batch from 400 to 1,000, or switch to on-demand shuttles.

- **Method:** Basic queueing logic (utilization ρ = arrival rate ÷ (servers × service rate), with waits rising sharply past roughly 0.85), Little's Law as a sanity check, and a simple Monte Carlo loop using observed arrival and service rates.
- **Problem solved:** Capacity decisions made by guesswork.

3.7 2035 Capacity Planner

- **What it does:** Converts headcount growth into required mess seats, beds, water, power, and shuttle trips for each year to 2035, with **trigger points** (for example, "when mess utilization passes 85%, start the expansion phase").
- **Method:** Headcount by year (from the institute's intake plan, or compound growth if unavailable) × measured per-capita use ÷ target utilization.
- **Problem solved:** Reactive, undated capacity planning. The roadmap becomes rule-based rather than date-based.

3.8 Unit Economics

Cost per student, carbon per student, and satisfaction score for each service, side by side. This gives the Director a single comparable view and underpins our cost-benefit analysis.

3.9 Student Services Assistant

An AI assistant that answers questions from institute circulars and FAQs, and routes anything it cannot answer to the right office as a ticket. It is the front door for students and parents to the same system.

4. Who Uses the System: User Flows

The design principle is that **each user sees only what they need**, mostly aggregated, and the Control Room never tracks individuals.

4.1 Student

1. Opens the app and sees today's menu and mess slot suggestions to avoid queues.
2. Opts in or out of a meal (this feeds the forecast).
3. Raises a hostel or facility complaint in one place, and sees status updates until it is closed.
4. Books a shuttle seat, or requests a trip in a slot where demand exists.
5. Asks the assistant a question about rules, schedules, or forms and gets an instant answer or a routed ticket.
6. Gives quick feedback (a one-tap rating), which feeds the satisfaction score.

4.2 Parent / Guardian

A deliberately limited view: official notices, campus transport and safety information, and a way to reach the right office through the assistant. No student-level tracking or personal data is exposed.

4.3 Faculty and Staff

Room and facility issues go through the same ticketing channel. Event and exam schedules entered by the academic office feed the master calendar, which in turn adjusts mess, energy, and transport forecasts.

4.4 Mess Manager

Each evening, receives tomorrow's forecast headcount, recommended cook quantity per item, and the ordering list. Records actual consumption and waste, which sharpens the next forecast.

4.5 Estates and Maintenance Team

Sees a prioritized ticket queue, scheduled preventive maintenance, and a repair-or-replace flag for repeat-failure assets. Uses the energy and water alerts to find leaks and abnormal loads.

4.6 Administration and Academic Office

Maintains the academic calendar (the master signal), receives the helpdesk SLA report, and sees which service requests repeat.

4.7 Director / Leadership

Uses Command Home for KPIs, the Simulator for "what if" decisions, the 2035 Planner for expansion triggers, and Unit Economics for budget conversations.

4.8 The Core Loop

Capture (check-ins, tickets, meters, requests) → **Forecast** (calendar-adjusted demand) → **Decide** (rules and recommendations) → **Act** (staff and students) → **Measure** (actual vs forecast, KPIs) → back to Capture. Every cycle adds data and improves the forecasts.

4.9 Resilience Mode Flow

When a disruption is flagged (road closure, power cut, connectivity loss), the system switches rules: the mess shifts to buffer-stock menus (cover days = disruption duration × daily demand), tickets are logged offline and synced later, and the shuttle plan is rerouted or paused.

5. Implementation Timeline with Data Gestation

A forecasting system is only as good as its data history. We are explicit about this: **some features cannot be switched on until enough data has been collected.** The timeline is therefore built around "data gestation" periods, with each advanced feature gated by a data readiness check.

5.1 Data Readiness: What Each Feature Needs

Feature	Minimum data before it is reliable	Method used while data is thin
Mess forecast (basic)	About 2-3 weeks of meal-wise headcount	Simple moving average
Mess forecast (day-of-week index)	About 4-6 weeks	Day-of-week factors on a moving average

Feature	Minimum data before it is reliable	Method used while data is thin
Event factors (exams, fests, breaks)	At least one full occurrence of each event type, ideally two	Manual judgement flagged as an assumption
Waste and inventory rules	About 4-6 weeks of waste and usage data	Conservative par levels
Helpdesk and MTBF analysis	6-12 months of ticket history (administration logs can shorten this)	Pareto on existing logs, preventive calendar from manufacturer schedules
Energy and water baselines	12 months to capture monsoon, winter, and summer. A 1-3 month baseline is only provisional	Bill history from estates plus a provisional baseline
Transport demand	4-6 weeks of requests (the survey provides a head start)	Survey-based demand histogram
What-if simulator	Observed arrival and service rates from 1-2 weeks of counter measurements	Rates measured by our own timing study
2035 Capacity Planner	Intake plan from administration plus per-capita use from the above	Compound growth assumption, clearly labeled
Unit economics	A full cost year for credible numbers	Budget data from administration

A practical rule: **start with moving averages and rule-based thresholds, and upgrade the method only when data supports it.** Report forecast error openly and show it falling as history accumulates.

5.2 Phased Timeline

Phase 0: Foundation and Baseline Study (Months 0-2, within the course project)

- Finalize the problem validation: observation, interviews, and a student survey.
- Request data from the mess, estates, and administration offices (these are the slow items).

- Run a short manual data collection: meal headcount, wastage estimate, queue times, a transport demand survey.
- Build the prototype for Demand and Mess and the mess Simulator using this data.
- **Gestation status:** data thin, so basic methods only, and all outputs labeled as prototypes.

Phase 1: Digital Basics and Data Collection (Year 1: 2026-27)

- Launch the unified helpdesk and ticketing so every complaint is logged.
- Launch meal check-in/opt-out and daily waste logging in the mess.
- Digitize the academic calendar as the master signal.
- Install basic sub-meters on hostels, the academic block, and the mess, and begin logging water use.
- Start transport requests and the student services assistant.
- **Gestation:** months 1-3 produce the first usable mess forecast (moving average). Months 4-6 add day-of-week and event factors. Full seasonal patterns need the complete year.
- **Gate to Phase 2:** mess forecast error within an agreed target (for example MAPE under 15%), and at least 6 months of ticket and meter data.

Phase 2: Integration and Prediction (Years 2-4: 2028-30)

- Integrate all modules into the Control Room, with the decision rules per module.
- Switch on preventive and predictive maintenance using the accumulated ticket history (MTBF, repair-vs-replace).
- Finalize energy and water baselines with a full year of data, and launch the anomaly alerts.
- Move transport to demand-based scheduling.
- Roll out Resilience Mode, tested through drills before monsoon.
- Pilot solar, rainwater harvesting, and the local sourcing and composting loop.
- Launch the 2035 Planner with real per-capita data.
- **Gate to Phase 3:** at least 24 months of clean data across modules, KPIs improving against the Phase 1 baseline, and a validated ROI on the pilots.

Phase 3: Optimization and Scale (Years 5-9: 2031-35)

- Expand the Simulator to a campus-wide flow model.

- Add advanced IoT and the full digital twin, but only where earlier data shows a clear payoff.
- Scale renewable energy and water systems toward carbon-neutral targets.
- Use the 2035 Planner trigger points to time expansions of the mess, hostels, and utilities.
- Review and refresh forecasts annually as the institute grows.

5.3 Summary Timeline

Period	Focus	Data status	Key outputs
Months 0-2	Baseline study and prototype	Manual samples, 1-3 weeks	Prototype screens, validated problem statement
Year 1 (2026-27)	Digital basics, start logging	Building history, 1-12 months	Ticketing, mess forecast v1, meters in place
Years 2-4 (2028-30)	Integration and prediction	1-3 years of history	Control Room live, predictive maintenance, resilience drills
Years 5-9 (2031-35)	Optimization and scale	4+ years of history	Digital twin, carbon-neutral initiatives, expansion by triggers

6. Risks and Safeguards

- **Thin data early on.** Mitigated by using simple methods first, stating assumptions, and gating features by data readiness.
- **Adoption.** If students do not use check-ins or ticketing, the data is poor. Keep inputs to one tap and show students the benefit (shorter queues, faster repairs).
- **Privacy.** The system works on aggregated data and never tracks individuals. Consent, anonymization, and compliance with India's DPDP Act are built in.
- **Infrastructure limits.** Offline-first design and Resilience Mode address connectivity and power issues.
- **Cost and skills.** Start with low-cost tools and phase hardware spending behind proven results.
- **Over-reliance on the model.** Managers keep override authority, and every recommendation shows the basis for it.

7. Expected Impact

Group	Expected benefit
Students	Shorter queues, faster repairs, fewer visits to offices, better-fitting transport
Faculty and staff	One system for requests, better scheduling visibility
Parents	Clear, reliable information without exposing student data
Institute	Lower waste and utility cost, evidence-based expansion, resilience to disruptions
Society and environment	Less food waste, lower carbon, support for local farmers and vendors

Targets for each benefit (for example, X% waste reduction, Y minutes saved in queues) should be set from our Phase 0 data and stated with assumptions.

8. Open Items for the Team

1. Confirm which screens will be built live (suggested: Demand and Mess plus the Simulator) and which will be wireframes with stated methods.
2. Send data requests to the mess, estates, and administration offices.
3. Divide the modules among team members so individual contributions are visible.
4. Add real numbers (costs, forecast error, queue times) from our fieldwork before final submission.